

COMP 495 Mentored Research Contract

STUDENT INFORMATION

Name William Keffer

RESEARCH INFORMATION

Project description:

Design and evaluate HeelCode, a course-configurable educational coding-agent harness for AI-allowed CS assignments. Instructor-authored, per-assignment policies govern student prompts, agent tool actions, and model outputs. A fine-tuned local encoder identifies observable help-seeking patterns: help scope, prior effort, and learning phase. It abstains when uncertain; explicit policy provides explainable scaffolds rather than misconduct claims. Compare rules, rubric-LLM, encoder, and hybrid systems; subject to IRB, run a matched crossover pilot with AI-free transfer tasks.

Meeting requirements:

Meet weekly for 60 minutes with mentor.
Bring agenda, reading notes, implementation/evaluation status, and decisions/risks.
Maintain a reproducible project log; complete milestones and submit material before pilot deployment.

Reading assignments:

Core readings:
CodeAid (Kazemitabaar et al., CHI 2024); CodeGuard (Raihan et al., Findings EACL 2026);
Cognitive Engagement Techniques (Kazemitabaar et al., IUI 2025); Generative AI Can Harm Learning (Bastani et al.); Policy to Prompting Instruction (Xiao et al., 2026); SWE-agent (Yang et al., NeurIPS 2024);
plus ModernBERT, DeBERTa-v3, AI Assessment Scale, ACM programming-assessment report,
and DeepSeek Harness docs.

Written assignments:

1. Annotated literature matrix and theory/novelty memo.
2. Policy specification, label codebook, and IRB/consent/data plan.
3. Evaluation protocol and preregistered model-promotion criteria.
4. Workshop-ready paper with reproducibility appendix and final report.

Software or hardware deliverables: (The deliverables should include the approximate scope and depth of work.)

1. Versioned assignment-policy schema and local policy-studio UI with validation and dry-run.
2. HeelCode intervention layer at prompt, tool-action, and output boundaries; explanations/retry and local audit events.
3. De-identified benchmark, annotation codebook, training/eval scripts, encoder model card, and calibration/abstention/error analyses.
4. Prototype demo; IRB-approved pilot build only if thresholds and approval are met.

Other assignments (e.g., presentations):

Midterm design review/demo; short poster or research-in-progress talk; final public demonstration and oral presentation. If a pilot runs, present results and lessons learned to participating instructor(s).

Assessment Plan:

Weekly progress reviewed with mentor. Completion judged by policy-studio and controller acceptance tests; labeled benchmark with agreement/error analysis; comparative held-out evaluation of rules, rubric-LLM, encoder, and hybrid systems; runtime/calibration/abstention results; pilot analysis if IRB-approved, otherwise usability and offline evaluation; quality of paper, reproducibility package, and final presentation.

Retention: This contract is to be retained for a minimum of four years.

Last update: 30 August 2018